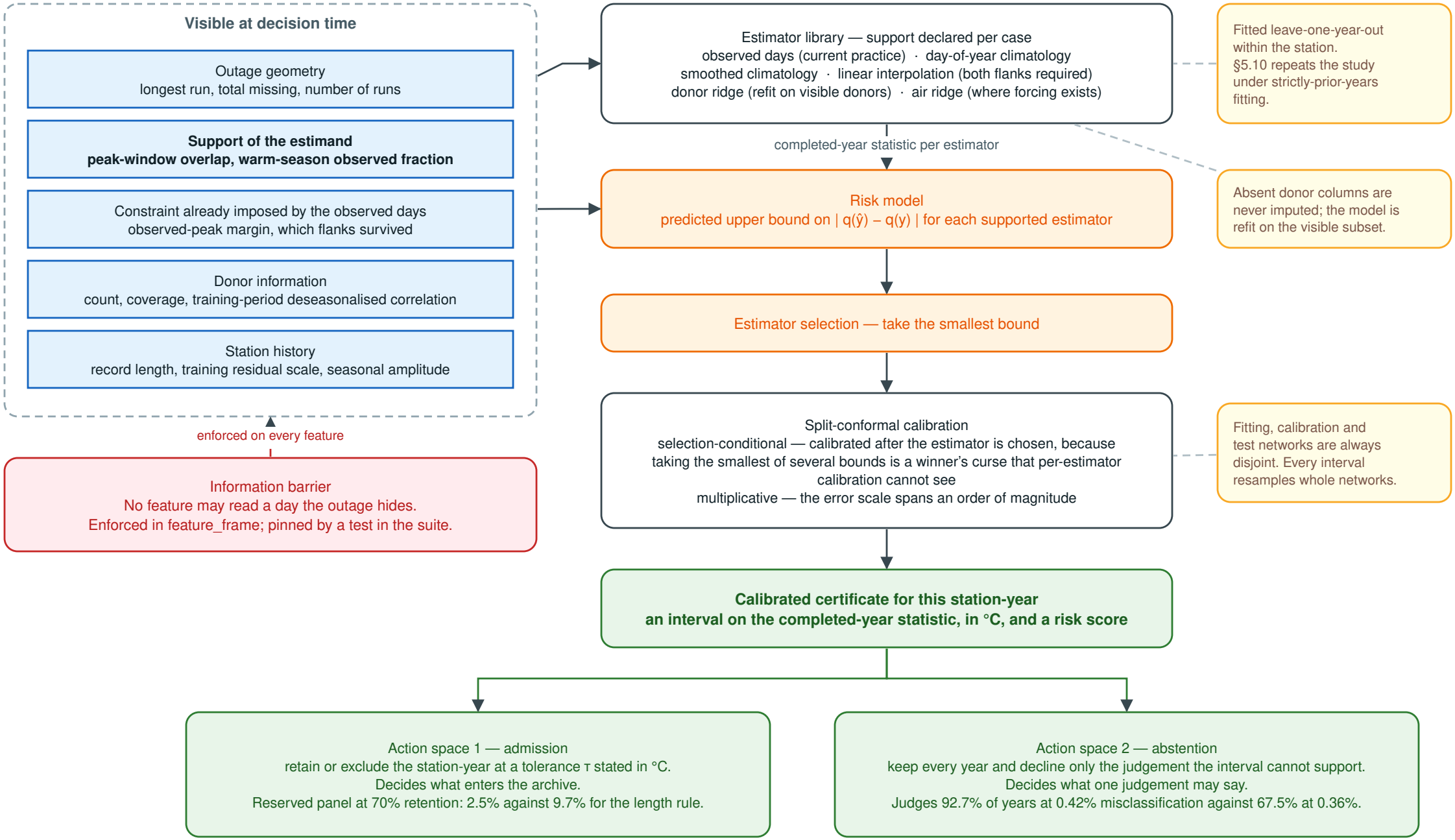
Visible at decision time
Outage geometry
longest run, total missing, number of runs
Support of the estimand
peak-window overlap, warm-season observed fraction
Constraint already imposed by the observed days
observed-peak margin, which flanks survived
Donor information
count, coverage, training-period deseasonalised correlation
Station history
record length, training residual scale, seasonal amplitude
enforced on every feature
Information barrier
No feature may read a day the outage hides.
Enforced in feature_frame; pinned by a test in the suite.
Estimator library — support declared per case
observed days (current practice) · day-of-year climatology
smoothed climatology · linear interpolation (both flanks required)
donor ridge (refit on visible donors) · air ridge (where forcing exists)
Fitted leave-one-year-out within the station. §5.10 repeats the study under strictly-prior-years fitting.
completed-year statistic per estimator
Risk model
predicted upper bound on | q(ŷ) − q(y) | for each supported estimator
Absent donor columns are never imputed; the model is refit on the visible subset.
Estimator selection — take the smallest bound
Split-conformal calibration
selection-conditional — calibrated after the estimator is chosen, because taking the smallest of several bounds is a winner's curse that per-estimator calibration cannot see
multiplicative — the error scale spans an order of magnitude
Fitting, calibration and test networks are always disjoint. Every interval resamples whole networks.
Calibrated certificate for this station-year
an interval on the completed-year statistic, in °C, and a risk score
Action space 1 — admission
retain or exclude the station-year at a tolerance τ stated in °C.
Decides what enters the archive.
Reserved panel at 70% retention: 2.5% against 9.7% for the length rule.
Action space 2 — abstention
keep every year and decline only the judgement the interval cannot support.
Decides what one judgement may say.
Judges 92.7% of years at 0.42% misclassification against 67.5% at 0.36%.